Chapter 6 : Combustion Systems

Fundamental Problem: inject fuel & oxidizer, mix them @ molecular level, have reactions to occur efficiently; do this without producing a normal shock & with a reasonable length of combustion chamber.

Engineers have been working on this since 60s. Only recently has this been successfully demonstrated with NASA-X43C

<u>Issues</u> : Mixing & chemistry

Define Damkoelher #

$$Da \equiv \frac{\text{mixing (fluid mech) timescale}}{\text{chem kinetic timescale}}$$

$$= \frac{\omega}{\chi}$$

ω : global chem rxn rate $[s^{-1}]$

$\chi \equiv$ scalar dissipation rate $[s^{-1}]$

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$$\omega \equiv A T^{\alpha} \exp\left(-\frac{E_a}{R_u T}\right)$$

A : Arrhenius const : might be a funct^n of concentration

α : temp exp. & typically between 0 – 2

E_a : activation energy.

$$\chi : 2D\,|\vec{\nabla} Z|^2$$

D : mass diffusion coeff.

$\vec{\nabla} Z$: gradient of mixture fraction

Look @ 2 extreme cases.

a) Mixing - limited.

Fuel & oxidizer react as soon as they mix → chem kin. time scales very short

$Da \Rightarrow$ very large.

thin reaction zone

b) kinetics limited.

Fuel & ox coexist, mix uniformly $\Rightarrow$ fluid mech time very short compared to rxn time

$\Rightarrow Da \ll 1$

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Mixture reacts homogenously, slowly converting reactants to products.

There are well validate theories for these two regimes.

Most devices fall under mixing-limited case.

Unfortunately supersonic combustion falls in between these limits & is hard to explain

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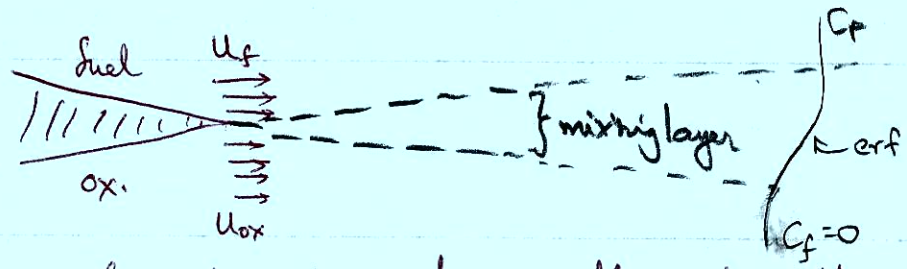
Mixing

Consider a mixing plate



If velocities are equal, there is no shear force & no large scale mixing. Mixing due to diffusion only.

Fick's law

$$\sigma_A = -D_{FA} \frac{\partial C_A}{\partial y}$$

D_{FA} : binary diffusion coeff.

$D_{FA} = D_{AF}$ (b/c it is binary)

σ_A : flux of A due to diffusion

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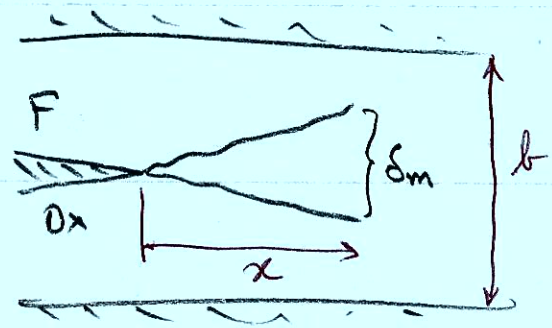
$\frac{\partial C_A}{\partial y}$ = max @ end of plate (where mixing starts)

$\sigma_A \equiv$ " " " " "

$\sigma_A \sim x^{-1/2}$ in this case.

(Note $\sigma_F \sim x^{-1/2}$ too).



Thickness of mixing layer

$$\delta_m = 8\left(\frac{D_{FA}x}{U_c}\right)^{1/2}$$

$$U_c = \left(\frac{U_F + U_O}{2}\right)$$, typically.

Set $\delta_m = b$
(shear layer occupying entire duct)

$$L_m = \frac{U_c b^2}{64 D_{FA}}$$

$L_m = 6$ ft for typical design.

Laminar; $\frac{L_m}{b} \sim 1{,}500$

Pure diffusion too slow!

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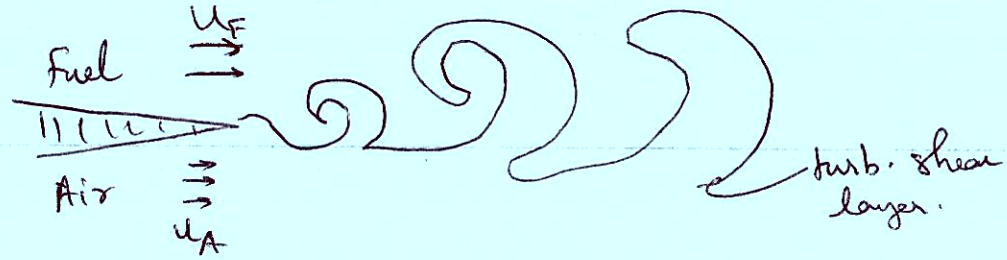
Case b)



- Due to difference in velocities, viscosity will cause momentum transport & formation of large ~~eat~~ vortical structures → Kelvin Helmholtz instabilities → very good for large-scale transport
- Large-scale structures both increase surface area & they also maintain high $\frac{\partial C_A}{\partial n}$
- Greatly increase σ_F, σ_A
- Since flow is turbulent, it is difficult to discuss in instantaneous sense. Instead look @ time average
- Time-averaged shear layer growth rate

$$\delta = C_\delta \left(\frac{1-r}{1+r}\right) x$$

$$r = \frac{u_2}{u_1} \qquad (u_2 < u_1)$$

$$C_\delta \approx 0.25 - 0.45$$

$$\frac{L_m}{b} \sim 50 \text{ for turb. mixing layer.}$$

Turb. shear layer ~~[illegible]~~ grows much faster than laminar, but still not fast enough

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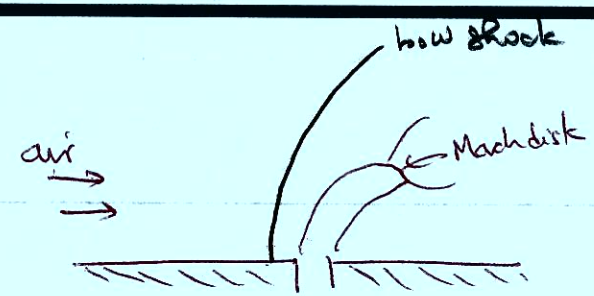
Normal injection



Benefit: very good initial mixing (near field)

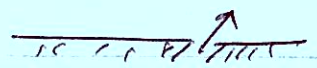
Problem: Normal shock causes loss of P_t.

Normal injection seems to only influence near-field mixing but far field still dominated by small-scale turbulence.

Normal injection has some benefits, but overall is probably negative.

Incoming fuel has large momentum which is wasted w/ normal injection

Angled injection



Component of fuel mom. is contributing to thrust.

Bow shock is weaker

Near-field mixing is worse.

HW 3 Ch 4: Prob 2, 4 using our η. & also derive eq^n 4.23
Ch 5 : # 4, 18

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